DE–3912 **101**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, MAY 2012.

DIGITAL COMPUTER ORGANIZATION

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Convert the following decimal number to the bases indicated.

   (i) 7852 to Octal

   (ii) 2008 to Hexadecimal

   (iii) 19875 to Binary (10)

   (b) Compare Boolean algebra with arithmetic and ordinary algebra. Discuss the basis theorems of Boolean algebra. (10)

2. (a) Explain MC Kluskey method. (10)

   (b) Describe the principle of Decoder. (10)

3. (a) Draw the circuit of a shift register and explain its working in detail. (10)

   (b) What is flip-flop? Discuss about D-flip-flop and SR-flip-flop. (10)

4. (a) What is ALU? Draw the block diagram of a 4 bit ALU. (10)

(b) Discuss the working of a parallel Binary Adder and sub tractor. (10)

5. (a) Discuss between static RAM and Dynamic RAM.(10)

(b) List out the advantages and disadvantages of semiconductor memories and magnetic memories. (10)

6. (a) Explain in detail about the memory addressing and encoding of information. (10)

(b) Explain the concept of segmentation in memory management. (10)

7. (a) Discuss about the one byte – address instructions and two byte – address instructions. (10)

(b) Explain in detail about Interrupts and operations of a computer during an interrupt. (10)

8. (a) Discuss in detail about program controlled I/O. (10)

(b) What is DMA? Explain the working principle of DMA with diagram. (10)

DE–3913 **102**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, MAY 2012.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2001 and 2002 batches)

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss in detail about revolution in computers and communications. (10)

   (b) Write shorts on "All Purpose Machine". (10)

2. (a) What are the types of application software? Explain each type in detail. (10)

   (b) Explain the features of word processing. (10)

3. (a) Write a note on intellectual property rights. (10)

   (b) List the various telephone related communications services, and explain each in detail. (10)

4. (a) Discuss web browsers and web search tools. (10)

   (b) Write notes on video and voice communication. (10)

5. (a) Discuss the factors affecting data transmission. (10)

(b) Write notes on work group computing. (10)

6. (a) List the advantages of networks and explain in detail about the information services. (10)

(b) Discuss about compression and decompression techniques. (10)

7. (a) What are the different types of database organization? Explain each one of them. (10)

(b) Discuss the advantages and disadvantages of DBMS. (10)

8. (a) Briefly explain about knowledge storage capacity. (10)

(b) Discuss about data normalization technique. (10)

DE–3914 **102**

DISTANCE EDUCATION

M.C.A. (N. S.) DEGREE EXAMINATION, MAY 2012.

PRINCIPLES OF INFORMATION TECHNOLOGY

(2001 and 2003 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain in detail about Telephone related communication services. (10)

   (b) Briefly discuss about EDI. (10)

2. (a) Discuss about presentation graphics software. (10)

   (b) Write short notes on Internet web browsers. (10)

3. (a) Explain about communication Networks in detail. (10)

   (b) Discuss about Outline Information services. (10)

4. (a) What is meant by bitwise operators? Give example. (10)

   (b) Explain in detail about structures. Write a program to display book details using structures. (10)

5. (a) How will you use strings in C? Explain in detail Multi dimensional arrays? Give example. (10)

   (b) Write short notes on arrays and vectors. (10)

6. (a) Discuss about pointers and functions with example. (13)

(b) Explain about Switch case constructions with example. (7)

7. (a) Write a Program for Matrix Multiplication. (10)

(b) Explain in detail about operators in C. (10)

8. (a) Briefly discuss about Input and Output streams. (10)

(b) Write a Program to design a Calculator using Switch case statement. (10)

DE–3915 **103**

DISTANCE EDUCATION

M.C.A. (N.S.) DEGREE EXAMINATION, MAY 2012.

C AND DATA STRUCTURE

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain any four control statements in C. (10)

   (b) Explain the two types of file organizations. (10)

2. (a) Write a procedure to convert infix to postfix expression. (10)

   (b) Describe the various operations on stacks and queues. (10)

3. (a) Explain the doubly linked list. (10)

   (b) Explain Huffman Algorithm. (10)

4. (a) Describe any three storage class specifies for variables? (10)

   (b) Illustrate the circular linked list with example. (10)

5. (a) Discuss about merge sort with an example. (10)

(b) Discuss about insertion sort with example. (10)

6. (a) Explain the sequential searching technique with an example. (10)

(b) Explain the various operations performed on an array. (10)

7. Explain the various tree traversal algorithms with example. (20)

8. (a) Explain the binary search with algorithm. (15)

(b) Define hashing techniques. (5)

DE–3916 **104**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

OFFICE AUTOMATION

(2011 onwards)

Time : Three hours Maximum : 100 marks

All questions carry equal marks.

Answer any FIVE questions.

($5 \times 20 = 100$)

1. (a) Briefly explain the features of Title bar, Status bar Menu bar. (15)

   (b) Explain in detail about WORD alignments. (5)

2. (a) Explain how to save a file and name the document. (5)

   (b) Discuss in detail about auto-formatting, table text, copying, moving and resizing pictures in word. (15)

3. (a) Explain the following :

   (i) Indenting.

   (ii) Spell checking. (10)

   (b) Discuss in detail about table and its options in WORD. (10)

4. (a) Explain in detail about building a worksheet and its functions. (10)

   (b) Explain the following with examples :

   (i) Autofill command

   (ii) Character functions. (10)

5. (a) Explain the various types of charts in Excel. (10)

(b) Explain how to create, format, print and chart in Excel. (10)

6. (a) Briefly explain the various data types in access. (10)

(b) How to add, insert and deleting records in access? (10)

7. (a) Discuss briefly about how to create the report in Ms-Access. (15)

(b) Briefly explain the features of access. (5)

8. (a) Explain the various steps involved in creating power point slides. (10)

(b) Briefly explain the various formatting options available in power point. (10)

DE–3917 **105**

DISTANCE EDUCATION

M.C.A. (N.S.) DEGREE EXAMINATION, MAY 2012.

DISCRETE MATHEMATICS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Show that $(P \vee Q) \wedge (\neg P \wedge (\neg P \wedge Q)) \Leftrightarrow (\neg P \wedge Q)$. (10)

   (b) Obtain the principal conjuctive normal form of $Q \wedge (P \vee \neg Q)$. (10)

2. (a) Show that $R \wedge (P \vee Q)$ is a valid conclusion from the premises $P \vee Q$, $Q \to R$, $P \to M$ and $\neg M$. (10)

   (b) Prove that :

   $(\exists x)(P(x) \wedge Q(x)) \Rightarrow (\exists x)P(x) \wedge (\exists x)Q(x)$. (10)

3. (a) Show that

   $A \times B = B \times A \Leftrightarrow (A = \phi) \vee (B = \phi) \vee (A = B)$. (10)

   (b) If $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ is the relation matrix of a relation R on the set $\{a, b, c\}$, find the relation matrices of $\widetilde{R}$, $R^2 = R \circ R$, $R^3 = R \circ R \circ R$ and $R \circ \widetilde{R}$. (10)

4. (a) If $f(x) = x + 2$, $g(x) = x - 2$ and $h(x) = 3x$ for $x \in R$, where R is the set of real numbers. Find $g \circ f$; $f \circ g$; $f \circ f$; $g \circ g$; $f \circ h$; $h \circ g$; $h \circ f$ and $f \circ h \circ g$. (10)

(b) Show that

$$f(A \cup B) = f(A) \cup f(B)$$

$$f(A \cap B) \subseteq f(A) \cap f(B).$$

Construct an example to show that in general it is not possible to replace $\subseteq$ by $=$ in the second relation. (10)

5. (a) Given the algebraic system $\langle N, + \rangle$ and $\langle Z_4, +_4 \rangle$, where N is the set of natural numbers and + is the operation of addition on N, show that there exists a homomorphism from $\langle N, + \rangle$ to $\langle Z_4, +_4 \rangle$. (10)

(b) If $S = \{a, b\}$, show that the semigroup $\langle S^s, \circ \rangle$ is not commutative. (5)

(c) If $V = \{a, b\}$ be an alphabet. Show that $\langle V^*, \circ, \wedge \rangle$ is an infinite monoid. (5)

6. (a) If $\langle G, * \rangle$ is an abelian group, then for all $a, b \in G$, show that $(a * b)^n = a^n * b^n$. (7)

(b) Show that if $\langle G, * \rangle$ is a cyclic group, then every subgroup of $\langle G, * \rangle$ must be cyclic. (7)

(c) State and prove Lagrange's theorem. (6)

7. (a) If $\langle G, * \rangle$ and $\langle H. \Delta \rangle$ be groups and $g : G \to H$ be a homomorphism. Then prove that the kernel of g is a normal subgroup. (9)

(b) Define integral domain. (3)

(c) Prove that every finite group of order n is isomorphic to a permutation group of degree n. (8)

8. (a) Define weighed graph. (3)

(b) Let A be the adjacency matrix of a digraph G. Then prove that the element in the ith row and j^{th} column of A^n (n is a non-negative integer) is equal to the number of paths of length n from the ith node to the jth node. (9)

(c) In a simple digraph $G = \langle V, E \rangle$, prove that every node of the digraph lies in exactly one strong component. (8)

DE–3918 **108**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

OBJECT ORIENTED PROGRAMMING AND C++

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Explain the basic concepts and evolution of object oriented programming in detail. (20)

2. (a) Discuss about standard I/O in C++. (5)

   (b) Explain stream classes for console I/O operations with a neat diagram. (15)

3. (a) Write a C++ program to add two time objects using binary operator overloading. (15)

   (b) What is member function? Give an example. (5)

4. (a) Discuss about function addresses with an example. (10)

   (b) Write the advantages and problems with reference in C++. (10)

5. (a) Explain how to create type conversion operator with an example. (10)

   (b) What is function overloading? Explain. (10)

6. (a) Write a C++ program to swap two numbers using pointers. (10)

(b) Discuss about single inheritance with an example. (10)

7. (a) Explain constructors and destructors with an example. (10)

(b) Discuss any one data conversion in C++. (10)

8. (a) What is pointer? Write its uses. (8)

(b) Write a C++ program to create an object and store it on Disk. (12)

DE–3919 **109**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, MAY 2012.

COMPUTER GRAPHICS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the various applications of computer graphics. (10)

   (b) Write in detail about Hardcopy devices. (10)

2. (a) Explain the midpoint circle drawing algorithm with example. (10)

   (b) Write short notes on

      (i) Boundary fill algorithm (5)

      (ii) Flood fill algorithm. (5)

3. (a) Explain in detail about the Sutherland-Hodman algorithm. (10)

   (b) Discuss in detail about Convex polygon clipping. (10)

4. (a) Discuss the composite transformation in two dimensional transformations. (10)

(b) Describe the matrix formulation of 2-D transformations; translation, rotation and scaling. (10)

5. (a) Write a short notes on

(i) General Pivot point rotation. (5)

(ii) General fixed point scaling (5)

(b) Explain about the different types of projection in 3-D. (10)

6. (a) Explain the general three dimensional rotations.(10)

(b) Explain the depth buffer method of hidden surface algorithm. (10)

7. (a) Define Clipping. Discuss Cohen-Sutherland algorithm for clipping. (10)

(b) Discuss in detail about the types of projection. (10)

8. (a) Explain the different models of user's interface design. (10)

(b) Explain in detail about the Components of User interface. (10)

DE–3920 **110**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

DESIGN AND ANALYSIS OF ALGORITHM

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Make a note on the development of a model. (10)

   (b) Explain the analysis of algorithm and its complexity. (10)

2. (a) Explain about linked list. (10)

   (b) Make a note on top down structured programming. (10)

3. (a) Make a note on program testing. (10)

   (b) What is implementation? Explain. (10)

4. (a) Explain Heap sort. (10)

   (b) Explain branching and bonding using travelling salesman problem. (10)

5. (a) Make a note on stack and queue. (10)

   (b) Explain the concept of backtracking. (10)

6. (a) What is recursion? Explain. (10)

   (b) Explain about Ackermann's function. (10)

7. (a) Explain Knight's Tour problem in chess. (10)

(b) Explain the Jeep problem. (10)

8. (a) What is Binary Tree Searching? Explain. (10)

(b) Explain Quick sort in detail. (10)

DE–3921 **111**

DISTANCE EDUCATION

M.C.A. (N.S.) DEGREE EXAMINATION, MAY 2012.

ACCOUNTING AND FINANCIAL MANAGEMENT

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

All questions carry equal marks.

1. What is a trading account? Mention the differences between the balance sheet.

2. The financial position of Mr. Saxish and Pujies on 1st January 2004 and December 31, 2004 was as follows.

Liabilities	1.1.04	31.12.04	Assets	1.1.04	31.12.04
	Rs.	Rs.		Rs.	Rs.
Current			Cash	4,000	3,600
liabilities	36,000	41,000	Debtors	35,000	38,400
Mr. Saxish loan	–	20,000	Stock	25,000	22,000
Loan from			Land	20,000	30,000
bank	30,000	25,000	Building	50,000	55,000
Capital	1,48,000	1,49,000	Machinery	80,000	86,000
	2,14,000	2,35,000		2,14,000	2,35,000

During the year the partners withdraw Rs. 26,000 for domestic expenses. The provision for depreciation against machinery as 1.1.2004 was Rs. 27,000 and on 31.12.2004 Rs. 36,000.

Prepare the cash flow statement for the year ended 31st December 1984.

3. The following information is extracted from the stores ledger.

Material X

Openings stock	Nil
Purchases :	
Jan 1	100 @ Rs. 1 per unit
Jan 20	100 @ Rs. 2 per unit
Issues :	
Jan 22	60 for Job W 16
Jan 23	60 for Job W 17

Complete the receipts and issues valuation by adopting the first in first out, last in first out and the weighted average method. Tabulate the values allocated to Job W 16, Job W 17 and the closing stock under the methods aforesaid.

4. P.S. Limited, manufactures of a special product, follows the policy of (E.O.Q) (Economic order quantity) for one of its components the components details are as follows.

Purchase price per component	Rs. 200
Cost of an order	Rs. 100

Annual cost of carrying one unit in inventory 10% of total cost of inventory and ordering per annum Rs. 4,000. The company has been offered a discount of 2% on the price of the component provided the lot size is 2000 components at a time. You are required to

(a) Compute the E.O.Q.

(b) Advise whether the quantity discount offer can be accepted.

(Assume that the inventory carrying cost does not variety accounting to discount policy)

(c) Would your advice differ if the company is offered 5% discount on a single order?

5. "The break even concept is fundamentally a static analysis". Explain the statement with reference to the Limitations of break-even concept.

6. Assuming that the cost structure and selling same in periods I and II find out.

   (a) Profit volume ratio

   (b) Break even point for sales

   (c) Profit when sales are Rs. 1,00,000

   (d) Sales required to earn a profit of Rs. 24,000 and

   (e) Safety margin in period II

   (f) Variable costs of the two periods.

Periods	Sales Rs.	Profit Rs.
I year	1,20,000	9,000
II year	1,40,000	13,000

7. From the following forecasts of income and expenditure prepare a cash budget for the three months commencing from 1st June, when the bank balance was 1,00,000.

	Sales	Purchases	Wages	Factory expenses	Administrative and selling expenses
April	80000	41000	5600	3900	10000
May	76500	40000	5400	4200	14000
June	78500	38500	5400	5100	15000
July	90000	37000	4800	5100	17000
August	95500	35000	4700	6000	13000

   (a) Assume that 50% are cash sales.

   (b) There is a two month credit period allowed to customers and received from suppliers

   (c) A sales commission of 5% on sales and the due two month to selling expenses

(d) Plant valued of Rs. 65,000 will be purchased and paid for in August

(e) The dividend for the last financial year of Rs. 15,000 will be paid in July

(f) Income-tax (advance) is to be paid in June Rs. 10,000

(g) Dividends from investments amounting to Rs. 1,000 are expected on 1st July

(h) Long in payment of wages 1/4 month

(i) Share call money for Rs. 30,000 is due from shareholders.

(Including a premium of Rs. 5,000 also) on 1st August.

8. A company produces a standard product. The estimated cost per unit is as follows :

Raw material Rs. 4; Direct labour Rs. 2, Variable overheads Rs. 5.

The semi-variable costs are : Indirect materials Rs. 235, Indirect labour Rs. 156, maintenance and repairs Rs. 570.

The variable costs per unit included in semi-variables are :

Indirect materials Rs. 0.05;

Indirect labour Rs. 0.08;

Maintenance and repairs Rs. 0.10.

The fixed costs are-factory Rs. 2,000;

Administration Rs. 3,000;

Selling and distribution Rs. 2,500.

The above costs are for 70% normal capacity producing 700 units. The selling price is Rs. 30 per units.

Prepare flexible budget for 60%, 80% and 100%. Normal capacities with the help of above.

DE–3922 **112**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

COMMUNICATION SKILLS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Discuss the various barriers to communication.

2. Explain the reading techniques.

3. Comment on the methods of asking questions.

4. Describe the process of group discussion.

5. How is body language a tool for personality identification?

6. What are the objectives of meetings?

7. How do you distinguish between downward and upward communication?

8. Write short notes on the following topics:

   (a) Agenda Writing (5)

   (b) Brain Storming (5)

   (c) Answering skills (5)

   (d) Personal style (5)

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DE–3923 **201**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, MAY 2012.

RESOURCE MANAGEMENT TECHNIQUES

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Solve by simplex method :

Maximize $z = 5x_1 + 3x_2$

Subject to :

$$x_1 + x_2 \le 2$$
$$5x_1 + 2x_2 \le 10$$
$$3x_1 + 8x_2 \le 12$$
$$x_1, x_2 \ge 0.$$

(b) Solve graphically the following L.P.P. :

Maximize $z = 3x_1 + 2x_2$

Subject to :

$$-2x_1 + x_2 \le 1$$
$$x_1 \le 2$$
$$x_1 + x_2 \le 3$$
$$x_1, x_2 \ge 0.$$

2. (a) What are the types of transportation problem? Distinguish transportation and assignment problem.

(b) Solve the following assignment problem.

	I	II	III	IV	V
A	6	5	8	11	16
B	1	13	16	1	10
C	16	11	8	8	8
D	9	14	12	10	16
E	10	13	11	8	16

3. (a) Discuss in detail about probabilistic models.

   (b) Write detailed notes on classification of queues and its types.

4. (a) Discuss in detail about the generation of random numbers.

   (b) Write down the algorithm for the single queuing system.

5. (a) Explain briefly about the replacement theory with various models.

   (b) Write detailed notes on finite and infinite delivery rates backordering.

6. (a) Write short notes on :

   (i) PERT

   (ii) ACTIVITY

   (iii) CRITICAL PATH.

   (b) Discuss the importance of dummies in network analysis.

7. Draw a network diagram and find the critical path for the following activities.

Activity	Time
1-2	4
1-3	1
2-4	1
3-4	1
3-5	6
4-9	5
5-6	4
5-7	8
6-8	1
7-8	2
8-10	5
9-10	7

8. (a) Discuss about the variable maintenance costs and variable money value.

(b) Explain briefly about the construction of time chart.

DE–3924 **202**

DISTANCE EDUCATION

M.C.A. (NS). DEGREE EXAMINATION, MAY 2012.

OPERATING SYSTEMS

(2001 onwards)

Time : Three hours Maximum : 100 marks.

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What is Operating System? Explain the concepts of Operating System. (10)

   (b) Describe the structure of OS. (10)

2. (a) Write a note on :

   (i) Multi programming. (5)

   (ii) System Calls. (5)

   (b) Discuss about Interprocess Communication. (10)

3. (a) List out the various process states and briefly explain with a state diagram. (10)

   (b) Explain Round robin scheduling and priority scheduling. (10)

4. (a) What is deadlock? How to prevent and avoid it? Explain. (12)

   (b) Write a note on I/O devices. (8)

5. (a) Discuss about :

(i) Race condition. (6)

(ii) Mutual Exclusion. (6)

(b) Explain about Interrupt handles. (8)

6. What is Paging? Explain the page replacement algorithms. (20)

7. (a) Explain Multiprogramming with fixed and variable partitions. (10)

(b) Discuss about file management. (10)

8. (a) Explain the concept of File System reliability. (8)

(b) Explain about directory structure. (4)

(c) Explain the mechanisms followed in file protection. (8)

DE–3925 **203**

DISTANCE EDUCATION

M.C.A.(N.S) DEGREE EXAMINATION, MAY 2012.

RDBMS

(2001 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain the desirable features of a database management system. (10)

   (b) Explain database schema and explain the design process of a database schema. (10)

2. (a) Write detailed notes on User Interfaces and explain the process involved in designing the user interface. (10)

   (b) Explain the Network Model in detail with example. (10)

3. (a) How the data is represented in relational model. Explain the semantic issues faced in RDBMS. (10)

   (b) Write detailed notes on architecture of oracle database. (10)

4. (a) Explain the table creation and modification process in oracle. (10)

(b) What is meant by client / server systems? Explain the role of client and server with suitable examples. (10)

5. (a) What are the uses of data manipulation language? Explain any five commands available in DML with suitable examples. (10)

(b) What are the tools available in Oracle environment? Explain them in detail. (10)

6. (a) Explain different joins with suitable examples. (10)

(b) Explain subqueries with suitable examples. (10)

7. (a) Write detailed notes on the following: (10)

(i) Schema

(ii) Union.

(b) Write notes on the following with examples. (10)

(i) Privileges

(ii) Objects.

8. (a) (i) How will you create a copy of an existing table? (5)

(ii) Write short notes on a pre compiler. (5)

(b) What is meant by index? How will you create index in oracle? How to use index? (10)

DE–3926 **204**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

UNIX AND SHELL PROGRAMMING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5×20 = 100)

1. (a) Explain devices and directories. (5)

   (b) Explain unix system application environment. (10)

   (c) Write short notes on program development. (5)

2. (a) What is structure? Explain the declaration of structure and accessing structure with suitable example. (10)

   (b) Write a program to display the following fibonacci series. 1, 1, 2, 3, 5, 8, 13. (10)

3. (a) Explain conditional and looping structures in shell programming. (10)

   (b) What is meant by shell? How to execute shell scripts. (5)

   (c) Write a short notes on .

      (i) Displaying current directory (2)

      (ii) Displaying the content of file (2)

      (iii) Renaming a file. (1)

4. (a) What are the uses of editing options in ex editor?(10)

(b) Explain about the directories and path names. (10)

5. (a) Explain different types of file operations with suitable examples. (10)

(b) Explain the following file name permissions with suitable example (10)

(i) Read

(ii) Write

(iii) Execute

6. Explain the following process. (20)

(a) PS

(b) no hub

(c) nice

(d) kill

(e) at

7. (a) Write short notes on the following :

(i) Unix programming tools. (5)

(ii) Command line arguments. (5)

(b) Explain control flow and relational operators. (10)

8. (a) Explain the text objects. (5)

(b) Explain error recovery. (5)

(c) Write notes on text mode and binary mode I/O functions. (10)

DE–3927 **205**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

COMPUTER NETWORKS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the uses of computer network. (5)

   (b) Give a brief note on OSI reference model. (15)

2. (a) Give a detail note on Telephone system. (10)

   (b) Write about communication satellites. (10)

3. (a) Explain the design issues of data link layer. (10)

   (b) Discuss about protocol specification and verification. (10)

4. (a) Give a brief note on routing algorithms. (10)

   (b) Discuss low network layer works in ATM Network. (10)

5. (a) Explain about ARP protocol. (10)

   (b) Explain the elements of transport protocol. (10)

6. (a) Give a brief note on TCP. (10)

   (b) Explain about traditional cryptography in detail. (10)

7. (a) Give short notes on digital signatures. (10)

(b) Explain the functions of DNS. (10)

8. (a) Explain electronic mail and its applications. (10)

(b) Give a brief note on multimedia packages. (10)

DE–3928 **208**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, MAY 2012.

SIMULATION AND MODELING

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. (5 × 20 = 100)

1. (a) Discuss the system modeling with an example. (10)

   (b) Distinguish the continuous system from the discrete system. Explain both the systems with examples. (5+5)

2. Discuss the procedures to implement the monete carlo method to simulate a systems. (20)

3. Explain the exponential growth model with an example. (20)

4. (a) Write an algorithm to generate uniform random numbers. (10)

   (b) What is arrival pattern? Explain with an example. (10)

5. (a) Discuss the various queuing disciplines. (10)

   (b) Explain the Rejection method congestion in systems. (10)

6. Distinguish discrete and continuous probability functions. (20)

7. (a) What is discrete event system simulation? (5)

(b) How do you simulate a system for inventory problem concerned with the purchase and sale of news paper? Explain. (15)

8. Discuss the simulation of manufacturing shop problem using GPSS language. (20)

DE–3929 **209**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

OBJECT ORIENTED ANALYSIS AND DESIGN

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss relationship among objects. (10)

   (b) Discuss about the Evolution of object model. (10)

2. (a) What are the four kinds of abstractions? Explain. (10)

   (b) What is encapsulation? Give an example. (10)

3. (a) Discuss the importance of proper classification. Briefly explain the different approaches used for identifying classes and objects. (10)

   (b) Explain the relationship that exit among objects.(10)

4. (a) What are the elements of a class diagram? Explain with example. (10)

   (b) Discuss any two advanced concepts of class diagram and their significance. (10)

5. (a) Explain the various object oriented analysis approaches. (10)

   (b) Write short notes on key abstraction mechanism.(10)

6. (a) What are the three roles played by an object? (10)

(b) What are the three approaches used to link objects to synchronization? Explain. (10)

7. Show the initial domain model for patient observation. Explain the same using object diagram. (20)

8. Write short notes on

(a) State diagram (10)

(b) Package diagram (10)

DE–3930 **210**

DISTANCE EDUCATION

M.C.A. (Non-Semester) DEGREE EXAMINATION, MAY 2012.

INTERNET PROGRAMMING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Differentiate Internet and Intranet. Explain its services.

   (b) Explain the features of CGI.

2. (a) What is Java applet? Explain in detail with an example.

   (b) Write a JavaScript to accept the details for students' record and to perform proper validation.

3. (a) Discuss in detail the features of VBScript.

   (b) Discuss the history of Internet.

4. (a) Discuss the software and protocols needed to design a web page.

   (b) What is Internet Explorer? Explain it and its application.

5. (a) Write short notes on :

   (i) Shokwave (5)

   (ii) Lingo. (5)

   (b) Write short notes on any one Internet Markup Language. (10)

6. Explain with suitable example the procedure of creating a custom Integrated Application with multiple protocols.

7. (a) Explain the method of creating and implementing the web design.

   (b) Explain the method of adding navigation bars to the web pages.

8. (a) Explain the web animation in detail.

   (b) Discuss the method of registering domain name for a web site.

DE–3931 **211**

DISTANCE EDUCATION

M.C.A. (Non-Semester) DEGREE EXAMINATION, MAY 2012.

SOFTWARE ENGINEERING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. Explain formal methods model. Describe about project management and tools. (20)

2. Discuss about making busy divisions. Explain risk monitoring and management in detail. (20)

3. Discuss on the following:

   (a) Project scheduling and tracking (10)

   (b) Configuration management. (10)

4. (a) Explain in detail about software prototyping. (10)

   (b) Explain about effective modular design. (10)

5. Explain in detail about:

   (a) Architectural design (10)

   (b) Procedural design. (10)

6. Write a brief note on:

(a) ISO 9000 quality standards (10)

(b) Software and formal technical reviews. (10)

7. Explain in detail about prototyping model and real model. (20)

8. Explain in detail about:

(a) White box testing and basis path testing (10)

(b) Human computer interface design. (10)

DE–3932 **212**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

VISUAL PROGRAMMING

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What are the various data types and windows resources? Explain. (8)

   (b) Discuss in detail about software Development Kit. (12)

2. Explain the following with an example each

   (a) Decision making statements. (10)

   (b) Looping. (10)

3. Explain in detail about Menus and Grid controls with an example. (20)

4. (a) Discuss about various recordset objects with an example. (10)

   (b) Compare and contrast between model and modeless dialog boxes. (10)

5. (a) Explain in detail about SDI and MDI. (15)

   (b) What are the various VC++ components? Explain.(5)

6. Explain the following :

(a) Splitter windows. (10)

(b) Multiple Document Inter face. (10)

7. (a) Discuss in detail about object Linking and Embedding. (15)

(b) Explain briefly about ODBC. (5)

8. (a) How to import VBX controls? Explain. (10)

(b) Write short notes on DLL. (10)

DE–3933 **301**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, MAY 2012.

SOFTWARE PROJECT MANAGEMENT

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What do you mean by project? Explain briefly about its characteristics. (5)

   (b) Discuss the various personality traits in detail. (15)

2. (a) Draw and explain the activity relationship diagram for a project of your choice. (10)

   (b) Briefly explain about the project checklist with examples. (10)

3. (a) Write a detailed note on project outlook. (10)

   (b) Discuss how tracking a project is done. Explain with examples. (10)

4. (a) Explain briefly about the product specifications. (10)

   (b) Why function test is needed? Explain the important objectives of function test plan. (10)

5. (a) Compare and contrast vendor and subcontractors.(5)

   (b) Describe the various performance measures in marketing issues. (15)

6. (a) How to implement the discipline in software project management? (10)

(b) Draw a project plan and activity responsibility matrix for a problem of your own assumption. (10)

7. (a) Explain briefly about the various approaches in planning a project. (10)

(b) Write a brief note on activity critical path. (10)

8. Discuss in detail about the important steps to be followed to yield a successful post project review. (20)

DE–3934 **302**

DISTANCE EDUCATION

M.C.A.(N.S) DEGREE EXAMINATION, MAY 2012.

CLIENT SERVER TECHNOLOGY

(2001 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What do you mean by client server? List and explain the advantages and disadvantages of client server. (10)

   (b) Discuss the various factors to differentiate the mainframe and client server systems. (10)

2. (a) Describe the role of client in client server environment in detail. (10)

   (b) Write a detailed note on Novell netware. (10)

3. (a) Explain briefly about OSI model and services provided by the layers with neat sketch. (10)

   (b) Write short notes on the following:

      (i) Interprocess Communication (5)

      (ii) Asynchronous Transfer Mode (ATM). (5)

4. (a) Why component is needed? List and explain the important properties of a component. (10)

   (b) Discuss in detail about Object Request Broker (ORB). (10)

5. (a) Describe the important factors that affecting client server application development. (10)

(b) Write a detailed note on Ellipse. (10)

6. (a) How client/server network differs from traditional centralized networking concept? Explain briefly. (10)

(b) Write short notes on Object Linking and Embedding. (10)

7. Write a brief note on server hardware architecture. (20)

8. Discuss any two server operating system in detail. (20)

DE–3935 **303**

DISTANCE EDUCATION

M.C.A.(NS) DEGREE EXAMINATION, MAY 2012.

PATTERN RECOGNITION AND IMAGE PROCESSING

(2001 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Define Histogram. Explain the Histogram equalization and histogram specification. (10)

   (b) What is image compression? Write short note on spatial and quantizing compression. (10)

2. (a) Discuss the grey level transformation. (10)

   (b) Define Edge detection. Explain the pyramid and crack edge detection. (10)

3. (a) Write short note on spatially dependent transformation. (10)

   (b) Explain the region growing, region split and merging techniques. (10)

4. (a) Describe the opening and closing morphological operations. (10)

   (b) Explain the image acquisition. (10)

5. (a) Define pattern recognition. Explain the various methodologies of pattern recognition. (10)

(b) Explain the linear decision functions. (10)

6. (a) Discuss the pattern classification by likelihood functions. (10)

(b) Explain the concepts of formal languages theory. (10)

7. (a) Explain the k-means algorithm. (12)

(b) Write a note on syntactic pattern description and recognition grammer. (8)

8. (a) How do you evaluate the clustering results? Explain. (10)

(b) Explain the cluster seeking algorithm. (10)

DE–3936 **304**

DISTANCE EDUCATION

M.C.A.(Non-Semester) DEGREE EXAMINATION, MAY 2012.

COMPILER DESIGN

(2001 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain different phases of a compiler. (10)

   (b) Explain Bootstrapping. (10)

2. With a neat diagram, explain the role played by lexical analysis. (20)

3. Write the algorithm for computing operator precedence relations and algorithm for operator precedence passing. (20)

4. Explain Data structures for symbol tables. (20)

5. Illustrate loop optimization with example. (20)

6. Construct SLR parsing table for the following grammar

   $E \rightarrow E+T/J$

   $T \rightarrow T*F/F$

   $F \rightarrow (E)/id.$ (20)

7. Elucidate in detail about the various peephole optimization techniques. (20)

8. Explain the various optimization techniques with examples. (20)

DE–3937 **305**

DISTANCE EDUCATION

M.C.A. (N.S) DEGREE EXAMINATION, MAY 2012.

MULTIMEDIA SYSTEMS

(2001 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

All questions carry equal marks.

1. (a) Define Multimedia? Give its various platforms.

   (b) Write note on Need, benefits and problems of Multimedia.

2. (a) What are the different types of Multimedia Development tools? Explain.

   (b) Explain any two commercial multimedia tools.

3. (a) Give the detail account in image and Graphics in Multimedia.

   (b) Write down the details about any two temporal media types.

4. (a) Describe the various simple video compression technique.

   (b) Discuss the objectives of JPEG image compression standards.

5. (a) Give your idea of object oriented concepts in Multimedia.

   (b) Write in detail about Transform and component classes.

6. (a) Write short notes on Media classes.

   (b) Give the details of Impact Disc family environment.

7. (a) Describe the multimedia system architecture and operations.

   (b) How is multimedia used in Network?

8. (a) How is multimedia applied in training and Education?

   (b) Write short notes on Media types.

DE–4372 **106**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

LAB I — DATA STRUCTURES USING 'C'

(Upto 2004)

(2001 and 2002 Batches)

Time : Three hours Maximum : 100 marks

Examiner has to choose the question for each candidate by lot system.

Each subdivision carries 50 marks.

Break-up of marks

Record Note Book	:	5
Algorithm/Flowchart	:	15
Program	:	35
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a 'C' program to implement all string operations using switch statement.

   (b) Write a 'C' program to copy the contents of one file to another file.

---------- Cut here ----------

2. (a) There are 10 students in a class. Their names and marks in three different subjects are given. If a student takes subjects are given. If a student takes more than 40 marks in each subject, then he is declared 'PASS' otherwise 'FAIL'. Write 'C' program to do the above using structure concept.

   (b) Write a 'C' program to sort a set of elements using Insertion sort.

3. (a) Write a 'C' program to find an element from the given of elements using binary search.

(b) Write a 'C' program to implement 'push' and 'pop' operations on stack.

---------- Cut here ----------

4. (a) A man is paid at the hourly rate of Rs 15/- per hour for the first 45 hours worked. There after, overtime is paid at 1.5 times the hourly rate for the next 25 hours and 2 times the hourly rate for these hours worked. Write a 'C' program to input the number of hours worker per week, Calculate and print his gross weakly wage.

(b) Write a 'C program to find the factorial of a given number using recursion.

---------- Cut here ----------

5. (a) Write a 'C' program to read 10 values to an array variable. Use pointers to locate and display each value.

(b) Write a 'C' program to perform matrix multiplication.

---------- Cut here ----------

6. (a) Write a 'C' program to implement insert and delete operations of lined list structure.

(b) Write a 'C' program to sort the set of elements using selection sort.

---------- Cut here ----------

7. (a) Write a 'C' program to find the roots of a quadratic equation using function definition $ax^2 + bx + c = 0$.

(b) Write a 'C' program to covert a given number into words for numbers to 5 (For eg. 1 to one, 2 to TWO and 5 to FIVE)

DE–4373 **107**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

COMPUTER LAB II — MS OFFICE

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Instruction :

Examiner has to choose the question for each candidate by lot system.

Each subdivision carries 50 marks.

Break up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1. (a) Using MS-Word, type the document in the newspaper format. Prepare a document with bullets and numbering.

(b) Create the following worksheet on MS Excel.

Reg No.	Name	Mark 1	Mark 2	Mark 3	Total	Average
–	–	–	–	–	–	–
–	–	–	–	–	–	–

(i) Using formula and function, find the total, average, maximum, minimum, total marks.

(ii) Sort the names in alphabetical order.

(iii) Create the bar chart for average mark with paper titles, legend and gridlines.

2. (a) Create a sales table with the following fields :

Field Name	Data type
Name	Text
Sex	Text
Sale code	Number
Sale amount	Number

Execute the following queries using the sales table.

(i) Display the records for sale amount>5000 and sex = "M".

(ii) Display the records for sex = "F" and sale amount between 5000 and 90000.

(iii) Sort the table in the descending order of sales amount.

(b) Using MS-Excel, create worksheet with the following details:

Name, description of the item, price of each item, quantity purchased, stock in hand, re-order level.

(i) Enter the five data in the above format.

(ii) Calculate amount = price × quantity purchased.

(iii) In table sort the field item wise.

(iv) Using OLE concept, exchange the worksheet into MS-Word table.

3. (a) Using MS-Access create address table with the following fields:

Field name	Data type
Name	Text
Age	Number
Sex	Text
Street	Text
City	Text
Pin	Number

(i) Add five records.

(ii) Display the fields name, street alone on the screen.

(iii) Display the records for Age>20.

(iv) Sort the table in the ascending order of name.

(b) Prepare three slides with a text and picture expressing the introduction of new product.

---------- Cut here ----------

4. (a) Type document regarding information technology and do the following :

(i) Find and replace the word using wild card and case matching.

(ii) Extract some paragraph to another file.

(iii) Do spell check in the document and add new word in the dictionary.

(iv) Change the font size and underline the paragraph.

(v) Use bullet and numbering.

(b) Create a worksheet in MS-Excel with the following details:

(i) Find the total, average of each student.

(ii) Find the highest and lowest scores in each subject.

(iii) Draw the bar chart as per the average mark.

(iv) Declare the result as 'PASS' if average is more than or equal to 50% otherwise 'FAIL'.

5. (a) Create Library table with the following fields:

Filed name	Type
Book number	Number
Title	Text
Author	Text
Price	Number
Published	Text
Branch	Text

Using MS-Access,

(i) Add five records using forms.

(ii) Display the records for title = "office automation" and author like "M*".

(iii) Display the records for book number = 5511 and branch = "CS".

(iv) Sort the table in the ascending order of book number.

(v) Display the records for publisher = "PHI' or publisher = 'BPB' and price between 300 and 500.

(b) An invoice table has the following details :

Name of the item, price of each items, number of quantity, item purchase, discount rate.

(i) Type the data in the above format using MS-Excel.

(ii) Find the total price, average using built in functions.

(iii) Sort the name in alphabetical order.

(iv) Draw a pie chart for total price.

6. (a) Prepare three slides with a text and picture expressing the introduction of new product.

(b) Using MS-Word, prepare your own biodata with the help of the template and use numbering and bullets where ever necessary.

---------- Cut here ----------

7. (a) Create student table with the following fields :

Filed name	Type
Reg. No	Number
Name	Text
Sex	Text
Age	Number
Degree	Text
Percentage	Number

Use MS-Access and execute the following queries:

(i) Add five records using forms.

(ii) Display the records for sex = "M" and degree = "MCA".

(iii) Display the records for sex = "F" and percentage between 70 and 80.

(iv) Display the records for degree = "MCA" and name like "M".

(v) Sort the table in the descending order of percentage.

(b) Prepare Pongal and Deepavali greeting cards with picture insertion and alignment. Write the procedure to take hard copy.

8. (a) Create worksheet in MS-Excel with the following details :

Name, description of the item, price of each item, quantity, purchased, stock in hand, re-order level.

(i) Enter the five data in the above format.

(ii) Calculate amount = price × quantity purchased.

(iii) Using OLE concept, exchange the worksheet into MS Word table.

(iv) In table, sort the field item wise.

(b) Create a table in MS-Word with the following data :

Account number, debit, credit, balance amount; enter the data, show with border shading.

---------- Cut here ----------

9. (a) Type the four columns and five rows, values in a document and convert into table format with headings. Use MS-Word.

(b) Create address table with following fields :

Field name	Data type
Name	Text
Age	Number
Sex	Text
Street	Text
City	Text
Pin	Number

(i) Using MS-Access, add five records.

(ii) Display the fields name, street above on the screen.

(iii) Display the records for age<20 and city = "Karaikudi".

(iv) Sort the table in the ascending order of name.

DE–4374 **113**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

LAB — III GRAPHICS USING C++

(2001–2002 batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

ONE question is to be given to each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C++ program to draw linear objects lines, rectangle using switch case option. First drawn the procedure and then implement the task.

   (b) Write a C++ program to draw a straight line using simple Breshnhams line algorithm.

---------- Cut here ----------

2. (a) Write a C++ program to draw a straight line using simple DDA technique.

   (b) Write procedure and a C++ program to draw nonlinear object ellipse.

3. (a) Write procedure and a C++ program to draw non-linear object circle and three concentric circle.

(b) Write a C++ program to show graphical transformation namely transformation of a two dimensional object.

---------- Cut here ----------

4. (a) Write a C++ program to draw non-linear objects arcs and curve.

(b) Write a C++ program to show graphical information scaling of a two dimensional object.

---------- Cut here ----------

5. (a) Write a C++ program to show graphical transformation rotation of a two dimensional object.

(b) Draw a cube and specify its co-ordinates.

---------- Cut here ----------

6. (a) Write a C++ program to show graphical transformation on three dimensional object.

(b) Write a C++ program to manipulate the graphical object using shading technique.

---------- Cut here ----------

7. (a) Draw a Circle . Divide into four quadratic and fillup different color for each quadrant.

(b) Draw any 3-D object and perform animation on it.

8. (a) Write a C++ program to draw a straight line using simple DDA technique.

(b) Draw a cube and specify its co-ordinates.

DE–4375 **114**

DISTANCE EDUCATION

M.C.A.(NS) DEGREE EXAMINATION, MAY 2012.

ALGORITHMS — Lab — IV

(2001 and 2002 Batch)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Break up of marks :
Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

ONE question is to be given to the Candidate by lot systems.

Each division carries 50 marks (Subdivision (a) and (b))

1. (a) Write a C program to perform Linear Search.

   (b) Write a C program to sort a given set of elements using Heap sort technique.

---------- Cut here ----------

2. (a) Write a C program to sort a given set of elements using Merge sort Technique.

   (b) Write a C program to check whether a Graph is connected using Depth first Technique.

3. (a) Write a C program to sort a given set of elements using selection sort.

(b) Write a C program to obtain the topological ordering of vertices in a given diagraph.

---------- Cut here ----------

4. (a) Write a C program to sort a given set of elements using Insertion sort.

(b) Write a C program to print all the nodes reachable from a given starting node in a diagraph using Breadth First search.

---------- Cut here ----------

5. (a) Write a C program to sort a given set of elements using Quick sort.

(b) Write a C program to print the triangle given below.

```
          1
       2     3
    4     5     6
   7  8     9     10
11    12    13    14  15
```

---------- Cut here ----------

6. (a) Write a C program to print all the nodes reachable from a given starting node in a given diagraph using Depth First search.

(b) Write a C Program to find the number of and sum of all integers greater that 100 and less than 200 that are divisible by a given integer x.

7. (a) Write a C program to Reverse the digits of the number. For example, 12345 should be printed as 54321.

(b) Write a C program to merge two Linked Lists.

---------- Cut here ----------

8. (a) Write a C program to implement a queue in which insertions, deletions and display can be performed.

(b) Write a C program to which accepts a graph as an adjacency matrix, and which performs depth first traversal on it and prints out the sequence of vertices visited.

DE–4376 **106**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

LAB I — DATA STRUCTURES USING 'C' & MS OFFICE

(2003 and 2004 Batches)

Time : Three hours Maximum : 100 marks

Examiner has to select TWO questions choosing One from C programs and another from MS-Office.

C PROGRAMS

1. Write a C program for linked list implementation of queue operations.

———————————————— cut here ————————————————

2. Write a C program to find the factorial of a given number using recursion.

———————————————— cut here ————————————————

3. Write a C program for matrix manipulation (Addition, Subtraction) using function.

———————————————— cut here ————————————————

4. Write a menu driven program in C to find an element using Linear and binary search methods.

———————————————— cut here ————————————————

5. Write a C program to sort a set of elements using selection sort.

MS-OFFICE PROGRAMS

6. Using MS-Word, prepare the first page of PROJECT NOTE BOOK with picture insertion and alignment.

---------- cut here ----------

7. Using MS-Word, prepare a document to advertise your company and tender notice using newspaper format.

---------- cut here ----------

8. Create the following worksheet on MSEXCEL

Reg. No.	Name	Mark 1	Mark 2	Mark 3	Total	Average
–	–	–	–	–	–	–
–	–	–	–	–	–	–

(a) Using formula and function, find the total, average, maximum, minimum total marks.

(b) Sort the names in alphabetical order.

(c) Create the bar chart for average mark with proper titles, legend and Gridlines.

---------- cut here ----------

9. Prepare three slides in powerpoint showing the features of MSOFFICE and also set timings to view it.

---------- cut here ----------

10. Prepare five slides in powerpoint with a text and picture expressing the courses offered in your university with animation effects.

DE–4377 **113**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012

LAB II : GRAPHICS USING C++ AND ALGORITHMS

(2003 and 2004 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Examiner should Select and Give Two questions to each candidate by Lot system.

1. Write a C++ program to draw a straight line using simple DDA technique.

---------- Cut here ----------

2. Write a C++ program to draw non-linear object circle and three concentric circle.

---------- Cut here ----------

3. Write a C++ program to show graphical transformation on three dimensional object.

4. Write a C++ program to manipulate the graphical object using SHADING technique.

---------- Cut here ----------

5. Write a C++ program for straight line exchange sorting.

---------- Cut here ----------

6. Write a c++ program to solve travelling saleman problem using Heuristic method.

---------- Cut here ----------

7. Write a C++ program to solve Tower of Honoi.

---------- Cut here ----------

8. Write a C++ program for recursion (Factorial and Fibonacci)

DE–4378 **206**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

LAB V – RDBMS

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Each question carries 100 marks. Examine has to choose and give ONE question for each candidate by lot system.

1. Create the following tables with 10 records for each.
Student (roll – no, name, category, district, states)
Student – rank (roll–no, marks, rank)
Write a PL/SQL procedure to generate a report to list of first 10 rankers from a particular district.

---------- Cut here ----------

2. An airline reservation database contains the reservation table and personal table.
reservation (flight – numbers, passengers–name, seat–no)
personal (passenger – name, sex, age, maritial – status, nationality)
Write a PL/SQL program to generate the following reports.

   (a) List the passengers name with seat numbers flight numberwise.

   (b) Total number of foreigners in a particular flight.

   (c) List out the passenger who have age above 70.

3. Create the following tables with 10 records each.
Book (accession–no, title, publisher, year, date-of purchase, status)
member (member-id, name, no. of-book-issue, max-limit)
Book–issue (accession-no, member-id, date-of-issue)

(a) Write a PL/SQL procedure to issue the book.

(b) Write a trigger to set the status of student to "black listed" if they have taken book but not returned even after one year.

---------- Cut here ----------

4. Create the following tables with 10 records each.
P-details (passenger–name, source, destination, time, date)
T-details (train–no, Train–name, source, destination, date, class, tariff)
Write a PL/SQL program to prepare a daily report with passenger details and tariffs for their reservation. The program should also check the availability of seats and availability of trains on a particular date.

---------- Cut here ----------

5. Write a PL/SQL program for stock–maintance system.
Item (item–code, item–name, 9th-in-stock, recorder–level)
Supplier (supplier–code, item–name, address, status)
can–supply (supplier–code, item–code)
Write a PL/SQL procedure to set the status of the supplier to "important" if the supplier can supply more than five times.

6. Write a PL/SQL program to calculate income tax of xyz private ltd employees based on the following conditions.

   (a) Gross pay upto 2,50,000 IT is 15% of the pay

   (b) Gross pay between 2,50,001 to 6,00,000 IT is 25% of the pay, above 2,50,000

   (c) Gross pay above 6,00,001 IT is 35% of the pay which is above 5,00,000.

---------- Cut here ----------

7. Write a program in PL/SQL to create a table having the fields as material number, material name, supplier and cost. The program should prepare a daily report with supplier and also generate monthly and yearly wise report.

---------- Cut here ----------

8. Write a program in PL/SQL to create a table having the attributes as doctor number, doctor name, experience, specialization program should display doctor information according to their specialization.

DE–4379 **207**

DISTANCE EDUCATION

M.C.A (NS) DEGREE EXAMINATION,
MAY 2012.

LAB VI — SHELL PROGRAMMING

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Examiner to select and give ONE question to each candidate.

Each sub division carries 50 marks.

Break-Up of Marks

Record Note Book	:	5
Algorithm	:	20
Program	:	30
Debugging & Execution	:	35
Result	:	10
Total	:	100

1. (a) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

   (i) file existence

   (ii) file readable

   (iii) file writable

   (iv) both readable and writable.

   (b) Write a menu driven shell program to copy, edit, rename and delete file.

2. (a) Write a menus driven shell program for the following :

(i) List of files

(ii) Processes of users

(iii) Today's date

(iv) Users of system

(v) Quit to UNIX.

(b) Write a menu driven shell program to perform the following tasks :

(i) Enter a sentence in file

(ii) Search a given whole word in an existing file

(iii) Quit.

---------- Cut here ----------

3. (a) Write a shell program using 3 arguments to take the pattern as well as input and output file names. If the pattern is found display "pattern found", else display "Error message". Also check if right number of arguments are entered.

(b) Write a shell program to prepare electric bill for domestic customers.

For first 100 units – Rs. 0.75/unit

For rent 100 units – Rs. 1.50/unit

Above 200 units – Rs. 3.00/unit

Prepare the bill for the following format :

Customer no : ————

Customer name : ————

Pre. reading : ————

Cur. reading : ————

Units consumed : ————

Charge : ————

4. (a) Write a shell program which accepts the name of a file from the standard input and then perform the following tests on it :

   (i) Enter the 5 names in a file

   (ii) Sort the name in existing file

   (iii) List unsorted and sorted file

   (iv) Quit.

   (b) Write a shell program to sum up the series :

$$\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\ldots+\frac{1}{10!}$$

---------- Cut here ----------

5. (a) Write a shell program to display the result 'PASS' or "FAIL" using the information given below.

   Student name, Student Reg.No, Mark 1, Mark 2, Mark 3, Mark 4, the minimum pass for each subject is 50.

   (b) Display the list of last 20 files present in the current directory. Also store this list in a file profile.

---------- Cut here ----------

6. (a) Write a shell program to convert all capital letters in a file to small letters.

   (b) Write a shell script which will receive either the file or the file name with its full path during execution. this script should obtain information about this file as given by Ls-1 display it in proper format.

---------- Cut here ----------

7. (a) Write a shell program for a file with each record containing name of city, name of state and name of country. Sort this file with country as a primary sort key and state as the secondary sort key.

   (b) Write the shall script which gets executed the moment the user logs in. It should display the message "Good morning" / "Good afternoon"/"Good evening". depending upon the time at which the user logs in.

8. (a) Merge the contents of the files file 1, file 2 and files sort them and display the sorted output on the screen page by page.

(b) Suppose a user has renamed some files in current. current directory using a command like mv file name file name $$. Write a shell script to research all such files and rename them such that they do not contain the shell PID.

DE–4380 **213**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

LAB VII – INTERNET PROGRAMMING

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Break Up of Marks

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

Examiner has to choose the question for each candidate by lot system. Each question carries 100 marks.

1. (a) Write a java script program to create an order form to select the house articles.

   (b) Write a HTML program to design and publish a web page for a college.

---------- Cut here ----------

2. (a) Write a HTML program to develop a web page for job recruitment agency in an IT industry.

   (b) Write a java program to set a color as the back ground.

3. (a) Write a VB script to do the following:

(i) Find the difference between the two dates

(ii) Find the age of a person by providing date of birth.

(iii) Display the current date.

(b) Write a java program using Applet to create the frames and its controls.

---------- Cut here ----------

4. (a) Write a menu driven java program to do the following database functions using database connectivity

(i) Insert

(ii) Delete and

(iii) Display records from a table.

(b) Using at least 20 HTML tags, create a screen with a string "web design".

---------- Cut here ----------

5. (a) Develop a picture gallery having atleast 3 pages. Each of them contains several pictures.

(b) Write a java program to display the dialogue and menu in Applet.

---------- Cut here ----------

6. (a) Write a java script to create an order form to select the house articles.

(b) Develop a single page advertisement for a shop to be opened shortly.

7. (a) Write a Java program using Applet to create the frames and its controls

(b) Create a web page in the format of front page of a newspaper using Text links. Align the text with colors.

---------- Cut here ----------

8. (a) Write a swing program to create buttons with

(i) Tool tip text

(ii) Image

(iii) Border

(iv) Short cut key.

(b) Write a Java script to create a color Palette and display the background in the color chosen from the Palette.

DE–4381 **214**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

LAB VIII — VISUAL PROGRAMMING

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Each candidate should select ONE question using Lot system.

1. (a) Write VB event procedure to find the sum of numbers from 1 to the selected value. Use a horizontal scroll bar to set the maximum value.

   (b) Write a VC++ program for creating SDI window and displaying help on right mouse click.

———————————— cut here ————————————

2. (a) Write a VB program to enter a file name in a Text box and search for the give file name in your system.

   (b) Write a VC++ program to create a document-view-frame-menu using Appwizard.

———————————— cut here ————————————

3. (a) Write a VB program to input two numbers and find its Addition, Subtraction, Multiplication, Division and Modulus.

   (b) Write a VC++ program for Mouse Events.

4. (a) Write a VB program to design a form and display the Height and Width accordingly when it is resized.

(b) Write a VC++ program to fill background of the client area with a bitmap.

---------- cut here ----------

5. (a) Write a VB program to move any object using timer control.

(b) Write a VC++ program to get the status of the shift and toggle key.

---------- cut here ----------

6. (a) Write a VB program to design a scientific calculator.

(b) Write a VC++ program to generate a status bar and show the status of caps lock, numlock and scroll lock in it.

---------- cut here ----------

7. (a) Create a VB application with a drive listbox, dirlist box.

(i) select exe file X execute it

(ii) select the picture and load it in to the form

(iii) filter a file according to the extension chosen.

(b) Write a VC++ program to create a list box in a window.

---------- cut here ----------

8. (a) Develop an application for railway reservation using VB.

(b) Write a VC++ program to create a dialog box in a window.

DE–4382 **206**

DISTANCE EDUCATION

M.C.A. (Non Semester) DEGREE EXAMINATION, MAY 2012.

LAB III — RDBMS AND SHELL PROGRAMMING

(2003 and 2004 Batches)

Time : Three hours Maximum : 100 marks

Each candidate should select a question using lot system.

Each subdivision carries 50 marks.

1. (a) A daily sales file contain record with the following fields :

Department number, date, item description, sale price for each item, quantity, cost of each item. Write a program using oracle to list all the input data and compute total amount of sales and profit for each sale. Accumulate and print total of sale and profit.

(b) Write a shell program which accepts the name of a file from the standard input and then performs the following tests on it :

(i) File existence

(ii) File readable

(iii) File writeable

(iv) Both readable and writeable.

2. (a) A salary statement contains Name, Basic pay, Allowance, Total deduction (includes IT), GPay, NPay, It is calculated on the basis of annual income under the following condition.

Annual salary	IT amount
< = 30000	NIl
> 30000 and < 55000	30% of the excess over the amount Rs. 30,000
> = 55000	50% of the excess over Rs. 55,000

Total deduction = deduction + income tax.

Allowance = 20% of BPay; GPay = BPay + Allowance, Deduction = 10% of BPay.

Write a program in SQL to prepare salary report for 5 employees.

(b) Write a menu driven shell program to

(i) Enter the sentence in file

(ii) Search a whole word in an existing file.

(iii) Quit.

---------- Cut here ----------

3. (a) Write a SQL program for canteen information system having two tables namely MENU and BILL table. The MENU table having the following fields. ITEM and UNIT RATE. Assume only items in this table. The BILL table having EMPNO, EMPNAME, DATE OF ISSUE, ITEM 1, TOKEN 1, (No. of tokens for items), RATE 1, ITEM 2, TOKEN 2, RATE 2, TOTAL AMOUNT. Prepare report with needed information.

(b) Write a shell program to prepare electric bill for domestic consumers. Prepare a bill format.

4. (a) Create a form for seat reservation problem in airlines reservation system and generate the reports for reservation details.

(b) Write a menu driven shell program for the following :

(i) List of files

(ii) Processes of users

(iii) Today's date

(iv) Users of system

(v) Quit.

---------- Cut here ----------

5. (a) A purchase table contains the following fields : Shop No, Item code, Data, Quantity and Price for each item. Write a PL/SQL program using oracle to find the item purchased in a particular date and amount.

(b) Write a shell program to display the result PASS or FAIL using the following information : Student Name, Reg. No., Mark 1, Mark 2, Mark 3, Mark 4. The minimum pass for each subject is 50.

---------- Cut here ----------

6. (a) The hospital billing system consists of the following fields, patient number, patient name, age, patient-type (IN/OUT). Consulting charge, blood test charge, Ex-ray charge, other test charges and total fee.

Write a menu driven program in SQL for the following conditions,

List of patients :

(i) Undergone blood test and charges.

(ii) Taken X-rays with charges.

(iii) Belong to In-patient category.

(iv) With total fee.

(b) Write a shell program for a file with each record containing name of city, name of state and name of country. How would you sort this file with country as the primary key and state as the secondary key?

7. (a) Write a program in SQL to compute the electric charges of electric units with the following condition.

For domestic	Rs. 0.55 for 1 unit
	Rs. 1.00 for exceed 100 units
For industry	Rs. 1.10 for 1 unit
	Rs. 1.40 for exceed 1000 units.

Prepare a neat report with needed details.

(b) (i) Merge the contents of the files, file 1, file 2 and file 3, sort them and display the sorted output on the screen page by page.

(ii) Write a shell script to display the list of last 20 files present in the current directory. Also store this list in a file profile.

---------- Cut here ----------

8. (a) Write a PL/SQL program and execute the same for a railway reservation system. Demonstrate reservation and cancellation operations.

(b) Create a shell program for a file contains records containing name of college, name of city and name of state. Sort this file with city as primary key and state as secondary key.

DE–4383 **213**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION,
MAY 2012.

LAB IV — INTERNET AND VISUAL PROGRAMMING

(2003 and 2004 Batches)

Time : Three hours Maximum : 100 marks

Examiner has to choose the question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a JavaScript to create a color palet and display the background in the color chosen from the palet.

   (b) Write a VB program to perform the following operation in a record of random Access file :

   (i) Insert

   (ii) Delete

   (iii) Search.

---------- Cut here ----------

2. (a) Write a VB Script to do the following :

   (i) Check the given password

   (ii) Change the existing password.

   (b) Write a Visual C++ program to handle Windows messages in MFC program.

3. (a) By using database connectivity (JDBC) insert and delete a records from a table.

(b) Write a Visual C++ program to create a list box in a window.

---------- Cut here ----------

4. (a) Write a Java program using Applet to display the dialogue and menu in applet.

(b) Write a visual C++ program to fill background of the client area with a bitmap.

---------- Cut here ----------

5. (a) Write a Java program using Applet to create the frames and its controls.

(b) Write a Visual C++ program to create a window of desired size using MFC.

---------- Cut here ----------

6. (a) Write a JavaScript to create a window by using the confirm message.

(b) Develop a web page for Job recruitment agency in an IT industry.

---------- Cut here ----------

7. (a) Write a JavaScript to create a order form to select the house articles.

(b) Develop a single page advertisement for a shop to be opened shortly.

---------- Cut here ----------

8. (a) By using database connectivity (JDBC) display the records in a table.

(b) Design and publish a web page for a college.

DE–4384 **306**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

LAB IX — MULTIMEDIA SYSTEMS

(2001 and 2002 batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Break up of marks
Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

Examiner has to choose the question for each candidate by lot system.

Each subdivision carries 50 marks.

1. (a) Write a C++ program to create a second image array model from an image, using grey-level transformation bunching technique. (50)

   (b) Design an image animation using flash. (50)

---------- Cut here ----------

2. (a) Write a C++ program for creating an image from another image, using automatic selection of grey level for splitting technique. Draw historgrams. (50)

   (b) Design a sales advertisement for any product using flash. (50)

3. (a) Write a C++ program to transform the grey-level using histogram equalization technique. (50)

(b) Using power point design the I.T. company advertisement. (50)

---------- Cut here ----------

4. (a) Write a C++ program for convolution using a template. (50)

(b) Prepare slides to show courses offered in a educational institution using power point presentation. (50)

---------- Cut here ----------

5. (a) Write a C++ program to compress the image using run-length encoding technique. (50)

(b) Do image animation using flash. (50)

---------- Cut here ----------

6. (a) Write a C++ program for 2 dimensional image transformations using computer graphics. (50)

(b) Design a sales advertisement for a product using powerpoint. (50)

---------- Cut here ----------

7. (a) Write a C++ program to find the edges of an image using sobel edge detection technique. (50)

(b) Design advertisement for a product using power point presentation. (50)

8. (a) Write a C++ program for 2 dimensional image animation using computer graphics. (50)

(b) Do an image animation using flash. (50)

DE–4385 **307**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

LAB X — COMPILER DESIGN

(2001 and 2002 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Candidates has to choose ONE question using lot system.

Each question carries 100 marks.

Break up of marks

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

1. Write a 'C' program which accepts $(0/1)^*011$ and draw the transition diagram for DFA.

---------- Cut here ----------

2. Given a grammar

$E \rightarrow E+E$

$E \rightarrow E*E$

$E \rightarrow (E)$

$E \rightarrow id$

Parse the string $(id_1 + id_2)^* id_3$ using shift reduce parsing algorithm.

3. Write a C program to accept a string and check whether it satisfies the following production rule (by using operator precedence grammar)

$S \rightarrow L = R$

$S \rightarrow R$

$L \rightarrow *R$

$L \rightarrow id$

$R \rightarrow L$

---------- Cut here ----------

4. Write a 'C' program to construct LR(0) items for the following production rule

$E \rightarrow E + T/T$

$T \rightarrow T * F/F$

$F \rightarrow (E)/id$

---------- Cut here ----------

5. Write a 'C' program to construct SLR parsing table for the following production rule

$S \rightarrow CC$

$C \rightarrow cC/d$

---------- Cut here ----------

6. Write a 'C' program to construct an LALR parsing table for the following grammar.

$S \rightarrow BB$

$B \rightarrow aB/b$

7. Write a 'C' program to find the right most and left most derivations for

$S \rightarrow iCtss'/a$

$S' \rightarrow eS/\in$

$C \rightarrow b$

Where $\in$ - empty string.

---------- Cut here ----------

8. Write a 'C' program to translate the expression $(a+b)*(b/c)$ into postfix notation, three address code, quadruples and indirect triples.

DE–4386 **306**

DISTANCE EDUCATION

M.C.A. (NS) DEGREE EXAMINATION, MAY 2012.

LAB V — MULTIMEDIA SYSTEM AND COMPILER DESIGN

(2003 and 2004 Batches)

(Upto 2004)

Time : Three hours Maximum : 100 marks

Examiner has to choose the question for each candidate by lot system. Each subdivision carries 50 marks.

1. (a) Create a photoshop image to show layer style.

   (b) Write a program in C to translate the expression its quadruples and triples (a + b)*(b – c)/(a + b).

---------- Cut here ----------

2. (a) Write a C++ program to transform the grey-level using Histogram Equalization technique.

   (b) Write a program in C to recognize identifier in the given source code.

---------- Cut here ----------

3. (a) Create a flash demo to animate your name.

   (b) Write a C program to covert infix to postfix notation. A + B * C/D.

4. (a) Design a sales advertisement for a product using flash.

(b) Write a program in C to recognize constant in the given source code.

---------- Cut here ----------

5. (a) Design an advertisement for a product using power point presentation.

(b) Write a C program which accepts $(1/0)1^{+}.10(1/0)^{*}$. And Draw the transition diagram for DFA.

---------- Cut here ----------

6. (a) Create a flash movie to add motion guide.

(b) Write a C program to implement lexical analyzer to remove the white spaces and to produce line number listing.

---------- Cut here ----------

7. (a) Create a flash program for various effects.

(b) Construct a c program to accept a string and check whether it satisfies the following production rule

(By using Operator Precedence Grammar)

$S \rightarrow AbB$

$A \rightarrow aBb$

$B \rightarrow a \mid Bba.$

8. (a) Prepare slides show for the course offered by an educational institution using Power point presentation.

(b) Write a C program to construct canonical LR parsing table for the following grammar.

$S \rightarrow XX, X \rightarrow yX / x$.